

Swetha Ramesh

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Personal Profile

Experienced software engineer with a master's in advanced computer science and two years of corporate experience specializing in data analysis, visualization, and software testing. Proficient in Python, SQL, Power BI and Tableau with a track record of enhancing team productivity through strong mentorship. I have the right to work in the UK.

EDUCATION

MSc in Advanced Computer Science | University of Leicester | Leicester, United Kingdom | 01.2023- 06.2024

- **Core Modules:** Big Data and Predictive Analytics(1st), Data Driven Service design (1st), Agile Cloud Automation, Cloud computing (1st)
- **Key Skills Developed:** Proficiency in machine learning, data visualization, and big data technologies, Expertise in Agile methodologies and cloud automation practices and Advanced user experience design and implementation.
- **Dissertation: Sentiment Analysis of UK Train Services Using Twitter Data (1st)**
- **Overall Grade:** Distinction

B.E in Information Science Engineering, 1st | Atria Institute of Technology, Bangalore, India | 07.2016- 07.2020

SKILLS

- **Programming Languages:** Python, SQL (MySQL, PostgreSQL)
- **Python Libraries:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, Scikit-learn, NLTK, TextBlob
- **Data Transformation & Visualization:** Power BI, Tableau, Excel.
- **Project & Workflow Management:** Jira, Flourish

WORK EXPERIENCE

Test Automation data Analyst, IBM

11.2021 to 11.2022

- Led **data-driven analysis** for AT&T's Enhanced Push-to-Talk (EPTT) web application, ensuring actionable insights for optimizing real-time communication platforms used by first responders.
- Extracted, transformed, and automated over 100+ data workflows using **Python, SQL**, Jira, Git, and Selenium. Managed and visualized datasets with **Excel** and **Power BI**, increasing operational efficiency by 30% and improving analytics-driven decisions by 25%.
- Designed and implemented CI/CD pipelines to streamline data processing workflows, reducing deployment times by 15% while accelerating analytics feedback cycles with enhanced data accuracy.
- Participated in code reviews and applied continuous integration practices to validate data pipelines, ensuring scalability and reliability while delivering actionable analytics for business intelligence.

Associate System Analyst - IBM

02.2021 to 11.2021

- Conducted data-driven analysis on test results using SQL, extracting key metrics that enhanced defect detection accuracy and reduced defect reoccurrence by 25%.
- Applied Power BI, Python, and SQL to visualize and interpret test data, uncovering critical insights into defect frequency, root cause analysis, and application stability for data-driven improvements.
- Collaborated closely with QA and development teams to refine data pipelines and processes, improving overall efficiency.
- Communicated technical findings effectively to business stakeholders, enabling alignment between technical and business objectives.

Intern, Atria IOT labs

10.2019 to 11.2019

- Developed an event reservation system using **RFID** tags with **python, SQL** and IoT technologies.
- Integrated IoT solutions using Python, improving system efficiency by 40% in a high-paced, collaborative team environment.

Intern, Hindustan Aeronautics Limited (HAL)-

04.2018 to 05.2018

- Collaborated with the HAL IT Department to revamp the CRM and Payroll System, focusing on data management and system optimization and reduced processing time by 20%. Enhanced online presence and ERP functionality by using C# and .Net framework and SQL for web development.

- Acquired practical proficiency in people analytics within HR and finance systems for over 2000 employees, honing problem solving and critical thinking capabilities.

CERTIFICATIONS

- **AWS** certified Cloud practitioner
- Power BI Essential Training by LinkedIn
- Data engineering Foundations by LinkedIn
- Microsoft certified **Azure fundamentals**
- Learning SQL foundations by LinkedIn
- Python for **Data Science**
- Python Basic issued by **Hacker Rank**
- IBM Agile Explore

PROJECTS

- **Dissertation: Sentiment Analysis of UK Train Services Using Twitter Data**
 - Implemented a project analyzing public **sentiments** toward UK Train Operating Companies (TOCs) using Twitter, employing **ETL**, NLP and machine learning techniques.
 - Leveraged advanced tools and methods including **Python** to extract, SQL(PostgreSQL) and pandas to transform and load the dataset, **Tableau dashboards** and Flourish to visualize the insights.
 - Established that Logistic Regression model leads with an accuracy of 76.32%, followed by SVM at 76.17%, with both models doing equally well across precision, recall, and F1-score criteria.
 - Created **data visualizations** (word clouds, regional sentiment maps) using Python, Tableau and Flourish to interpret sentiment trends effectively followed by a user evaluation study.
- **Portfolio Website**
 - Developed and hosted a responsive portfolio website via GitHub and deployed on Netlify to showcase my projects, experience, and skills. Built with **HTML**, **CSS**, **JavaScript**, and **Bootstrap**. Leveraged ChatGPT for prompt engineering to streamline troubleshooting and code optimization.
 - Technologies: HTML, CSS, JavaScript, Bootstrap, GitHub, Netlify, ChatGPT for Prompt.
- **Neighborhood Property Price Analysis and Predictive Modeling for Strategic Insights:**
 - Designed visualizations by employing **Matplotlib**, **Plotly**, and **Power BI** to analyze large-scale customer data to extract trends and insights. Discovered detailed patterns like how property prices vary by neighborhood and change over time.
 - Compared a basic linear regression model with a more advanced one that included methods like **K-means clustering** and **random forest regression**.
 - Found that the advanced model performed better with an 87% accuracy and additionally translated the complex datasets into clear, actionable insights, proving the ability to support strategic decision-making—skills applicable to optimizing marketing campaigns and evaluating performance metrics.
- **Distributed Whiteboard Application on Azure:**
 - Led the collaborative development and deployment of a real-time, distributed whiteboard application on Azure, showcasing expertise in cloud architecture, application development, and seamless integration with **Azure Web PubSub**.
 - Transformed a stand-alone application into a scalable and collaborative platform, while identifying areas for future enhancements in data persistence and scalability.
- **Detection of tomato growth state and surveillance system using computer vision and IoT:**
 - Implemented an IoT and **computer vision**-powered **data analytics and visualization** system for monitoring tomato growth stages.
 - Optimized irrigation practices using a Raspberry Pi, DHT sensor for environmental data collection, and **Python's OpenCV** for image processing and analysis and MySQL for database management, enabling data-driven decision-making through the correlation and modeling of visual and sensor data.

OTHER EXPERIENCE

- Student Mirror board student representative at the University of Leicester.
- Café Assistant at Sainsbury's Café.
- Fluent in 5 Languages including English and Hindi.
- Volunteered 20+ hours per month at a non-profit café in Leicester, supporting community outreach and event planning.